NAME

mbclean − Tool to automatically flag bad beams in swath sonar bathymetry data.

VERSION

Version 5.0

SYNOPSIS

mbclean
[--deviation-max=max {-Amax}
--depth-range=low/high {-Blow/high}
--max-slope=slope/unit {-Cslope/unit}
--distance-range=min/max {-Dmin/max}
--max-acrosstrack=max_acrosstrack {-Emax_acrosstrack}
--format=format {-Fformat}
--fraction-range=fraction_low/fraction_high {-Gfraction_low/fraction_high}
--input=infile {-Iinfile}
--range-min=range_min {-Krange_min}
--lonflip=lonflip {-Llonflip}
--mode=mode {-Mmode}
--ping-deviation-tolerance=tolerance {-Ntolerance}
--speed-range=min_speed/max_speed {-Pmin_speed/max_speed}
--zap-rails=backup_dist {-Qbackup_dist}
--max-heading-rate=maxheadingrate {-Rmaxheadingrate}
--spike-max=spike_slope/mode/format {-Sspike_slope/mode/format}
--timestamp-tolerance=tolerance {-Ttolerance}
--min-good-beams=num_good_min {-Unum_good_min}
--bounds=west/east/south/north {-Wwest/east/south/north}
--zap-beams=beamsleft/beamsright {-Xbeamsleft/beamsright}
--flag-distance-angle=distanceleft/distanceright[/mode] {-Ydistanceleft/distanceright[/mode]}
--zero-position {-Z}
--verbose {-V}
--help {-H}]
mbclean −−input=infile −−cube−spacing=dx[/dy]
[--cube-region=west/east/south/north
--cube-flag=hypothesis|outlier|both
--cube-flag-threshold=threshold
--cube-flag-nadir=angle/threshold[/shallow|both]
--cube-flag-strength-max=strength
--cube-prediction-topography
--cube-* options of **mbgrid** −F9 (see below)
--format=format {-Fformat} --lonflip=lonflip {-Llonflip} --verbose {-V}]

DESCRIPTION

mbclean identifies and flags artifacts in swath sonar bathymetry data. Several algorithms are available for identifying artifacts; multiple algorithms can be applied in a single pass. The most commonly used approach is to identify artifacts based on excessive bathymetric slopes. Spikes where an excessive slope occurs before and reverses after a beam can also be removed. If desired, **mbclean** will also flag beams associated with "rails" where outer beams have smaller acrosstrack distances than more inner beams (**−Q** option). Soundings can also be compared with a CUBE surface built from all of the input data, as in **mbgrid** −F9 (**−−cube−spacing** option, see CUBE FLAGGING below). Low and high bounds on acceptable depth values can be set; depth values outside the acceptable range will be flagged. The acceptable depth ranges can either be absolute (**−B** option), relative to the local median depth (**−A** option) or defined by low and high fractions of the local

median depth (**−G** option). A set number of outer beams can also be flagged.

The order in which the flagging algorithms are applied is as follows:

1. Flag specified numbers of outer beams (**−X** option).
2. Flag outer beams and/or unflag inner beams by acrosstrack distance (**−Y** option).
3. Flag all beams in pings outside specified acceptable speed range (**−P** option).
4. Flag all beams in pings outside specified acceptable position bounds (**−W** option).
5. Flag all beams in pings with zero longitude and latitude values (**−Z** option).
6. Flag soundings outside specified acceptable depth range (**−B** option).
7. Flag soundings with ranges less than specified minimum value (**−B** option).
8. Flag pings with excessive heading change rate (**−R** option).
9. Zap "rails" (**−Q** option).
10. Flag soundings with acrosstrack distances greater than specified maximum value (**−E** option).
11. Flag soundings outside acceptable depth range using fractions of local median depth (**−G** option).
12. Flag soundings outside acceptable depth range using deviation from local median depth (**−A** option).
13. Flag soundings associated with spikes (**−S** option).
14. Flag soundings associated with excessive slopes (**−C** option or default).
15. Flag all soundings in pings with too few good soundings (**−U** option).
16. Flag all soundings in pings that on average deviate too much from the pings before and after (**−N** option).

This program flags beams by outputting the flags as edit events to an "edit save file", like that produced by **mbedit**. If an "edit save file" (named by adding a ".esf" suffix to the input swath filename) already exists, the edits are read in and applied before the **mbclean** flagging algorithms are used. Once generated, the edit events can be applied to the data using the program **mbprocess**, which outputs a processed swath data file. The **mbprocess** program is also used to merge edited navigation, recalculate bathymetry, and apply other corrections to swath bathymetry data.

CUBE FLAGGING

With the **−−cube−spacing** option **mbclean** builds a CUBE (Combined Uncertainty and Bathymetric Estimator; Calder and Mayer, 2003) surface from all of the input data and flags the soundings that do not belong to it. This is the "data cleaning" use of CUBE described in the CUBE User's Manual (Calder and Wells, 2007): soundings that do not belong to the depth hypotheses selected by CUBE are flagged. The CUBE flagging cannot be combined with the other tests of **mbclean** in one run; run **mbclean** once with the other tests and once with CUBE flagging.

The CUBE grid is the one **mbgrid** −F9 builds with the same region (**−−cube−region**, **mbgrid** −R), the same

node spacing in meters (**−−cube−spacing**, **mbgrid** −E*dx/dy*/meters) and the same **−−cube−*** options on a geographic grid (no **−J**), so the surface the soundings are judged against can be seen by running **mbgrid**. The data are read as **mbgrid** reads them: processed files where the datalist asks for them (**$PROCESSED**), raw files otherwise, and alternative navigation where the datalist gives it. In addition the current edits of each raw file (its edit save file) are applied, so soundings already flagged are left out. If a processed file is older than its raw file, parameter file or edit save file, a warning asks for **mbprocess** to be run first. **mbgrid** sees these edits only in processed files: for edited files without processed versions, run **mbprocess** before comparing with **mbgrid**. The edit save file is registered in the parameter file only when edits are added, so a run that flags nothing leaves the processed files up to date.

The data are read twice: the first time to build the CUBE grid, the second time to compare every unflagged sounding with it. A median predicted surface (**−−cube−prediction=median**) adds one more read. Only the CUBE grid is kept in memory. Each sounding is offered to the nodes as during the insertion into the grid (influence radius, capture distance, and the blunder test and slope correction of a predicted surface); the nearest of these nodes holding data decides. There the following criteria are applied (by default only **outlier** and **blunder**, see **−−cube−flag**):

hypothesis
: The sounding is flagged if the depth hypothesis it belongs to is not the one selected for the node. As in the CUBE estimator, a sounding belongs to the hypothesis with the smallest normalized error |z − d| / sqrt(v_d + v), where d and v_d are the predicted depth and variance of the hypothesis and v the variance of the sounding propagated to the node. This criterion never flags soundings at nodes with a single hypothesis.

outlier
: The sounding is flagged if its normalized error from the selected hypothesis exceeds a threshold. By default the threshold is the one at which the CUBE monitor rejects a single sample, (hˆ2 − 2 ln B) / (2 h) with h from **−−cube−est−offset** and B from **−−cube−bayes−factor**: 2.50 with the defaults. With **−−cube−flag−nadir**, soundings in the nadir sector have their own threshold and, by default, are flagged only when they are shallower than the hypothesis (see below).

blunder
: With a predicted surface and the blunder test on, soundings that the blunder test rejects at the deciding node are always flagged.

Soundings that reach no node holding data (for example near the edges of the grid) are not tested and are counted as such. Only flags are added; nothing is unflagged. The flags are appended to the edit save file of the raw file as filter flags, as for the other **mbclean** tests; they cannot be told apart from other filter flags once **mbprocess** has applied them.

Deep water multibeam data can show a centre-beam artefact: near nadir, bottom detections that are too shallow, by a fraction of a percent to a few percent of the depth, often over several adjacent beams. These soundings are one-sided (almost all shallower than the seafloor) and many are small, so a symmetric threshold that catches them also flags many valid soundings in the outer beams and on slopes. **−−cube−flag−nadir** gives the soundings in a sector around nadir their own outlier threshold, applied by default only to soundings shallower than the selected hypothesis; the soundings outside the sector keep the threshold of **−−cube−flag−threshold**. The beam angle is the angle from the vertical between the sonar and the sounding, from the horizontal distance between the sonar position and the sounding and the depth of the sounding below the sonar. The CUBE surface itself is the same with and without this option.

The test depends on the sounding uncertainties as much as the CUBE surface does. The propagated variance v grows with the horizontal uncertainty of the sounding relative to the node spacing, so with the default IHO S-44 Order 2 model, whose horizontal uncertainty is 20 m plus 10% of the depth, soundings in shallow water or on a fine grid are rarely flagged. Choose **−−cube−order** or the **constant** or **depth** models (**−−cube−model**) to match the survey, as described in the **mbgrid** manual page.

The node spacing must resolve the terrain. Each sounding is compared with the estimate at a node that can be up to 0.7 node spacings away, and the slope between them is not part of CUBE's variance propagation: with tight vertical uncertainties and a coarse grid over rough or sloping seafloor, many valid soundings exceed the outlier threshold. In a test on a deep water EM122 survey (1400 to 5300 m), the default model at 200 m flagged 0.004% of the soundings, most of them gross blunders, whereas a vertical uncertainty of 0.14% of depth flagged 28% at 200 m and 4.7% at 50 m. Check the number of flagged soundings, and look at them in **mbeditviz**, before applying the edits with **mbprocess**.

MB-SYSTEM AUTHORSHIP

David W. Caress
Monterey Bay Aquarium Research Institute
Dale N. Chayes
Center for Coastal and Ocean Mapping
University of New Hampshire
Christian do Santos Ferreira
MARUM - Center for Marine Environmental Sciences
University of Bremen

ALSO CONTRIBUTED TO THIS PROGRAM

Alberto Malinverno
Lamont-Doherty Earth Observatory
Dana Yoerger
Woods Hole Oceanographic Institution

OPTIONS

−−deviation−max=max −Amax

This option sets the range of acceptable depth values relative to the local median depth. The median depth is obtained from the current ping and the pings immediately before and after that ping. If a depth value deviates from the median depth by more than *max*, then it will be flagged. No deviation from the median depth checking is done if the **−A** option is not used.

−−depth−range=low/high −Blow/high

This option sets the range of acceptable depth values. If a depth value is less than *low* or more than *high* then it will be flagged. No depth range checking is done if the **−B** option is not used.

−−max−slope=slope/unit −Cslope/unit

The value *slope* is the maximum acceptable slope. Beams associated with excessive slopes will be flagged or removed according to the operational mode specified using the **−M** option. This method will be used if no other algorithms are specified; if other algorithms are specified but **−C** is not used then no slope checking will occur. *unit* is optional and specifies the unit of *slope*, 0 (default) indicates the slope is in tangents, 1 slope is in radians, 2 slope is in degrees. Default: *slope* = 1.0

−−distance−range=min/max −Dmin/max

Sets the minimum and maximum allowed distances between beams used for some of the flagging algorithms. Both values are expressed in terms of fractions of the local median depth. Thus, **−D***0.01/0.25* will translate, if the local median depth is 1000 meters, to a minimum distance of 10 meters and a maximum distance of 250 meters. The *min* value sets the minimum distance between beams required for an excessive slope to be used to flag bad beams. The navigation and heading of the ship are used to calculate the locations of beams. Ship turns often cause beams of adjacent pings to overlap, causing the distances between these beams to become quite small. This can, in turn, magnify noise in the bathymetry data to produce slope estimates which are excessively large. The *max* value sets the maximum distance between the current beam and other beams for those beams to be used in evaluating the current beam. For instance, only beams within the maximum distance are used to calculate the local median depth, and only beams within the maximum distance are used to check

for excessive slopes. Default: *min/max* = 0.01/0.25.

−−max−acrosstrack=max_acrosstrack −Emax_acrosstrack
This option sets the maximum acceptable acrosstrack distance for a sounding. If the absolute value of a beam's acrosstrack distance exceeds *max*, then it will be flagged. No acrosstrack distance checking is done if the **−E** option is not used. Default: *max* = 120.0 meters.

−−format=format −Fformat
Sets the data format used if the input is read from stdin or from a file. If *format* < 0, then the input file specified with the **−I** option will actually contain a list of input swath sonar data files. This program uses the **MBIO** library and will read or write any swath sonar format supported by **MBIO**. A list of the swath sonar data formats currently supported by **MBIO** and their identifier values is given in the **MBIO** manual page. Default: *format* = 11.

−−fraction−range=fraction_low/fraction_high −Gfraction_low/fraction_high
This option sets the range of acceptable depth values relative to low and high fractions of the local median depth. The median depth is obtained from the current ping and the pings immediately before and after that ping. If a depth value is less than *fraction_low* times the median depth (e.g. *fraction_low* = 0.5 means one half the median depth) or more than *fraction_high* times the median depth then it will be flagged. No fractional depth range checking is done if the **−G** option is not used.

−−help −H
This "help" flag cause the program to print out a description of its operation and then exit immediately.

−−input=infile −Iinfile
Sets the input filename. If *format* > 0 (set with the **−F** option) then the swath sonar data contained in *infile* is read and processed. If *format* < 0, then *infile* is assumed to be an ascii file containing a list of the input swath sonar data files to be processed and their formats. The program will read and process the data in each one of these files. Each input file will have an associated output file with either the ".sga" or ".aga" suffix. In the *infile* file, each data file should be followed by a data format identifier, e.g.:

```
datafile1 11
datafile2 24
```

This program uses the **MBIO** library and will read or write any swath sonar format supported by **MBIO**. A list of the swath sonar data formats currently supported by **MBIO** and their identifier values is given in the **MBIO** manual page. Default: *infile* = "datalist.mb-1".

−−range−min=range_min −Krange_min
This option causes all unflagged beams with ranges less than *range_min* to be flagged as bad. The value *range_min* is specified in meters.

−−lonflip=lonflip −Llonflip
Sets the range of the longitude values used. If *lonflip*=−1 then the longitude values will be in the range from −360 to 0 degrees. If *lonflip*=0 then the longitude values will be in the range from −180 to 180 degrees. If *lonflip*=1 then the longitude values will be in the range from 0 to 360 degrees. Default: *lonflip* = 0.

−−mode=mode −Mmode
Sets the manner in which bad beams identified by excessive slope are handled.

mode = 1:	Flags one beam associated with each outlier slope. The flagged beam is the one furthest from the local median depth.
mode = 2:	Flags both beams associated with each outlier slope. Default: *mode* = 1.

−−ping−deviation−tolerance=tolerance −Ntolerance
When specified **mbclean** calculates the rms deviation between each ping and the pings before and after. When the rms deviation exceeds the specified threshold, then all unflagged valid soundings in that

ping are flagged as bad. This option is generally used to detect and remove data pings affected by interference from other active sonars.

−−speed−range=min_speed/max_speed −Pmin_speed/max_speed
This option causes **mbclean** to flag as bad all beams in pings associated with platform speed outside the acceptable range from *speed_low* to *speed_high*. The speed values are specified in km/hour.

−−zap−rails=backup_dist −Qbackup_dist
This flag causes **mbclean** to search for bad "rails" in the swath sonar swath; the "rails" refer to groups of outer beams which have crosstrack distances (and depths) much less than they should have. These are identified when one or more outer beams lies more than *backup* meters inboard of a more inner beam; all beams meeting this criteria are flagged.

−−max−heading−rate=maxheadingrate −Rmaxheadingrate
The value *maxheadingrate* is the maximum acceptable rate of change in heading in degrees/second. All soundings associated with pings for which the heading was changing at a greater rate will be flagged.

−−spike−max=spike_slope/mode/format −Sspike_slope/mode/format
The value *slope* is the maximum acceptable spike slope. If the slope from the preceding beam to this beam exceeds this value, and the slope from this beam to subsequent beam exceeds this value but with an opposite sign this beam is considered a spike and will be flagged or removed according to the operational mode specified using the **−M** option. Acrosstrack slopes are determined by the preceding and subsequent beams in the same ping. Alongtrack slopes are determined from the same beam in the previous and subsequent pings. Alongtrack are fairly sensitive to the minimum distance **−D** option, which will normally need to be set less to a very small value for alongtrack slopes to be detected. There is no test that alongtrack distances are all in the same direction.

If *mode* is 1 (default) only acrosstrack spikes are detected. If *mode* is 2 only alongtrack spikes are detected. If *mode* is 3 both along track and across track slopes are checked.

unit is optional and specifies the unit of *slope*, 0 (default) indicates the slope is in tangents, 1 slope is in radians, 2 slope is in degrees. A beam is not considered a spike if either the preceding or subsequent beam has already been flagged. Default: *slope* = 1.0

−−timestamp−tolerance=tolerance −Ttolerance
If requested this option will reset the timestamps of edit events from an existing esf file to exactly match the timestamps of the survey pings. The *tolerance* value sets how close timestamps must be in seconds to be considered a match. This option handles the case where survey data have been processed using non-**MB-System** software and a user is extracting the edits from one set of files with **mbgetesf** and then applying them to another using **mbprocess**.

−−min−good−beams=num_good_min −Unum_good_min
This flag causes **mbclean** to search for port or starboard halves of pings which contain fewer than *nmin* good bathymetry values. All bathymetry values in the affected half-pings are flagged.

−−verbose −V
Normally, **mbclean** works "silently" without outputting anything to the stderr stream. If the **−V** flag is given, then **mbclean** works in a "verbose" mode and outputs the program version being used, all error status messages, and the number of beams flagged as bad.

−−bounds=west/east/south/north −Wwest/east/south/north
This option causes **mbclean** to flag as bad all beams in pings with navigation outside the specified acceptable bounds.

−−zap−beams=beamsleft/beamsright −Xbeamsleft/beamsright
If this option is used, the outermost *zap_beams* at both ends of the swath are flagged as bad; this is useful if the outer beams are known to be unreliable. Default: *zap_beams* = 0.

−−flag−distance−angle=distanceleft/distanceright[/mode] −Ydistanceleft/distanceright[/mode]
This option causes **mbclean** to flag and/or unflag beams according to acrosstrack distance. If *mode*=1 or is not specified, then all beams outside the specified range (acrosstrack distances more to port/left than *distanceleft* or more to starboard/right than *distanceright*) will be flagged as bad. If *mode*=2, then all beams inside the specified range (acrosstrack distances more to starboard/right than *distanceleft* and more to port/left than *distanceright*) will be flagged as good. The distances are defined in meters, and distances to port of nadir are negative. If *mode*=3, then all beams outside the specified range of beam angles (angles from vertical more to port/left than *distanceleft* or more to starboard/right than *distanceright*) will be flagged as bad; this is the angular equivalent of *mode*=1. If *mode*=4, then all beams inside the specified range of beam angles (angles from vertical more to starboard/right than *distanceleft* and more to port/left than *distanceright*) will be flagged as good; this is the angular equivalent of *mode*=2. For *mode*=3 and *mode*=4, *distanceleft* and *distanceright* are beam angles expressed in degrees rather than acrosstrack distances in meters, with angles to port of vertical negative.

−−zero−position −Z
This option causes **mbclean** to flag as bad all beams in pings with zero longitude and latitude values.

−−cube−spacing=dx[/dy]
Turns on CUBE flagging (see CUBE FLAGGING) with a CUBE grid of node spacing *dx* meters east and *dy* meters north (default *dy* = *dx*). CUBE flagging cannot be combined with the other tests.

−−cube−region=west/east/south/north
Sets the bounds of the CUBE grid in degrees. By default the bounds of the input data are used (from the ".inf" files, which are created if needed), as in **mbgrid**.

−−cube−flag=hypothesis|outlier|both
Selects the CUBE flagging criteria (see CUBE FLAGGING). Default: outlier. The **hypothesis** criterion, the literal rule of the CUBE User's Manual, also flags valid soundings where CUBE has split a sloping or rough seafloor into several hypotheses at a node; use it only with a node spacing that resolves the terrain, and preferably with **−−cube−flag−strength−max**.

−−cube−flag−threshold=threshold
Sets the normalized error from the selected hypothesis above which the **outlier** criterion flags a sounding. Default: the single sample outlier threshold of the CUBE monitor, 2.50 with the default **−−cube−est−offset** and **−−cube−bayes−factor**.

−−cube−flag−nadir=angle/threshold[/shallow|both]
Gives the soundings whose beam angle from the vertical is less than *angle* degrees on either side of nadir their own **outlier** threshold *threshold* (a normalized error, see CUBE FLAGGING). With **shallow** (the default) only soundings shallower than the selected hypothesis by more than *threshold* are flagged there; with **both**, soundings deeper by more than *threshold* are flagged too. Soundings outside the sector keep the threshold of **−−cube−flag−threshold**. Needs the **outlier** criterion. With **−V** the beam angle is added at the end of the lines listing the flagged soundings. Default: off.

−−cube−flag−strength−max=strength
Applies the **hypothesis** criterion only at nodes whose hypothesis strength (0 for a certain node, up to 5 for an ambiguous one) is at most *strength*, leaving soundings at ambiguous nodes for inspection, for example in **mbeditviz**. Default: 5 (no limit).

−−cube−prediction−topography
The predicted surface grid file (−−cube−prediction=*gridfile*) holds heights (positive up), as a grid made with **mbgrid** −A2 or a global relief model, instead of depths (positive down).

−−cube−*
The CUBE algorithm, uncertainty model and predicted surface options of **mbgrid** −F9 are accepted with the same meanings and defaults: −−cube−bayes−factor, −−cube−blunder, −−cube−capture−min, −−cube−capture−scale, −−cube−context, −−cube−depth−reference, −−cube−dist−exp,

−−cube−est−offset, −−cube−extractor, −−cube−horiz−scale, −−cube−iho−ccom, −−cube−iho−edition, −−cube−influence−scale, −−cube−median−length, −−cube−memory−limit, −−cube−model, −−cube−model−a, −−cube−model−b, −−cube−model−c, −−cube−order, −−cube−prediction, −−cube−prediction−hypotheses, −−cube−prediction−uncertainty, −−cube−quotient−limit, −−cube−runlength, −−cube−sigma−h and −−cube−sigma−v. See the **mbgrid** manual page. A predicted surface grid file (−−cube−prediction=*gridfile*) must be a geographic grid of depths (positive down), or of heights (positive up) with **−−cube−prediction−topography**. If more of the soundings compared with it have the opposite sign than the same sign, **mbclean** stops after the first read of the data without flagging anything. With the likelihood and posterior selection rules, the −−cube−context distances must reach at least one node spacing (−−cube−spacing); otherwise the rules fall back to the prior rule and a warning is printed.

EXAMPLES

Suppose one wishes to do a first pass edit of six Simrad EM300 files in the processing format (format 57). A datalist referencing these six files exists as the file datalist.mb-1 and has the contents:

```
0001_20020424_212920.mb57 57
0002_20020425_011607.mb57 57
0003_20020425_022926.mb57 57
0004_20020425_024336.mb57 57
0005_20020425_034057.mb57 57
0006_20020425_045013.mb57 57
```

Use the following to flag any beams which deviate by more than 20% from the local median depth or which produce a slope greater than 3.5 (74 degrees):

```
mbclean −Idatalist.mb-1 \
        -M1 −C3.5 −D0.01/0.20 \
        -G0.80/1.20
```

The program will output flagging statistics for each file and give totals at the end. If the **−V** option is specified, **mbclean** will also output information for each beam that is flagged. Here is an example of the nonverbose output:

```
Processing 0001_20020424_212920.mb57
908 bathymetry data records processed
0 outer beams zapped
0 beams zapped for too few good beams in ping
0 beams out of acceptable depth range
64 beams out of acceptable fractional depth range
0 beams exceed acceptable deviation from median depth
0 bad rail beams identified
1601 excessive slopes identified        0 excessive spikes identified
1665 beams flagged
0 beams unflagged

Processing 0002_20020425_011607.mb57
259 bathymetry data records processed
0 outer beams zapped
0 beams zapped for too few good beams in ping
0 beams out of acceptable depth range
0 beams out of acceptable fractional depth range
0 beams exceed acceptable deviation from median depth
0 bad rail beams identified
242 excessive slopes identified        0 excessive spikes identified
242 beams flagged
```

```
0 beams unflagged

Processing 0003_20020425_022926.mb57
65 bathymetry data records processed
0 outer beams zapped
0 beams zapped for too few good beams in ping
0 beams out of acceptable depth range
9 beams out of acceptable fractional depth range
0 beams exceed acceptable deviation from median depth
0 bad rail beams identified
497 excessive slopes identified          0 excessive spikes identified
506 beams flagged
0 beams unflagged

Processing 0004_20020425_024336.mb57
410 bathymetry data records processed
0 outer beams zapped
0 beams zapped for too few good beams in ping
0 beams out of acceptable depth range
0 beams out of acceptable fractional depth range
0 beams exceed acceptable deviation from median depth
0 bad rail beams identified
148 excessive slopes identified          0 excessive spikes identified
148 beams flagged
0 beams unflagged

Processing 0005_20020425_034057.mb57
252 bathymetry data records processed
0 outer beams zapped
0 beams zapped for too few good beams in ping
0 beams out of acceptable depth range
0 beams out of acceptable fractional depth range
0 beams exceed acceptable deviation from median depth
0 bad rail beams identified
100 excessive slopes identified          0 excessive spikes identified
100 beams flagged
0 beams unflagged

Processing 0006_20020425_045013.mb57
562 bathymetry data records processed
0 outer beams zapped
0 beams zapped for too few good beams in ping
0 beams out of acceptable depth range
0 beams out of acceptable fractional depth range
0 beams exceed acceptable deviation from median depth
0 bad rail beams identified
41 excessive slopes identified          0 excessive spikes identified
41 beams flagged
0 beams unflagged

MBclean Processing Totals:
------------------------
6 total swath data files processed
```

2456 total bathymetry data records processed
0 total beams flagged in old esf files
0 total beams unflagged in old esf files
0 total beams zeroed in old esf files
0 total outer beams zapped
0 total beams zapped for too few good beams in ping
0 total beams out of acceptable depth range
73 total beams out of acceptable fractional depth range
0 total beams exceed acceptable deviation from median depth
0 total bad rail beams identified
2629 total excessive slopes identified 0 total excessive spikes identified
2702 total beams flagged
0 total beams unflagged

To flag the soundings of the same data that do not belong to a CUBE surface with 50 m node spacing, for a deep water survey from a surface vessel (IHO S-44 Order 2, the default):

```
mbclean −Idatalist.mb-1 −−cube−spacing=50
```

and to see the surface the soundings were compared with:

```
mbgrid −Idatalist.mb-1 −Ocube −F9 −E50/50/meters −A1 −M
```

Here **mbgrid** must use the same region as **mbclean**, which by default is the bounds of the data for both programs. If the processed files are out of date, run **mbprocess** first.

In deep water the horizontal uncertainties of the IHO S-44 orders (Order 2: 20 m plus 10% of the depth; Orders 1a and 1b: 5 m plus 5%) leave only gross blunders flagged. To flag more, describe the sounding uncertainties with the **depth** model instead, here one standard deviation of 0.3% of the depth vertically and 1 m horizontally, and raise the outlier threshold (**−−cube−order=special** then only sets the TVU bound of the influence radius):

```
mbclean −Idatalist.mb-1 −−cube−spacing=50 −−cube−order=special \
  −−cube−model=depth −−cube−model−a=0.13 −−cube−model−b=0.003 \
  −−cube−sigma−h=1 −−cube−model−c=0 −−cube−flag−threshold=3.5
```

On the EM122 survey described under CUBE FLAGGING this flagged 0.76% of the soundings. Compared with a manual cleaning of the same survey in **mbeditviz**, 34% of these flags agreed with manual flags and they covered 30% of the manual flags; most of the manual flags not found were small offsets near nadir. A 100 m node spacing flagged more valid soundings on steep slopes. Tune the vertical uncertainty and the threshold together (their product, here about 1% of the depth, sets the cutoff) and inspect the result in **mbeditviz** before running **mbprocess**.

To aim at the centre-beam artefact instead, flag soundings within 10 degrees of nadir that are more than 1.5 normalized errors shallower than the CUBE surface, and elsewhere only those more than 5 normalized errors away:

```
mbclean −Idatalist.mb-1 −−cube−spacing=50 −−cube−order=special \
  −−cube−flag−nadir=10/1.5 −−cube−flag−threshold=5
```

On the same EM122 survey, preprocessed and corrected for sound speed, this flagged 10,231 soundings (0.44%), of which 54% agreed with manual flags, covering 27% of the manual flags; a single threshold of 2.5 with the same grid and model flagged 19,759 soundings for 29% agreement and 29% coverage. The sector

and the thresholds were chosen on this survey; check them on other systems and depths.

Surveys that combine steep slopes and flat plains may need more than one set of parameters. On the EM122 survey above, the centre-beam artefact was 1 to 2% of the depth on the slopes but only about 0.2% (around 10 m at 5000 m) on the abyssal plain, inside the scatter of the valid soundings there. A nadir threshold low enough for the plain flagged many valid soundings on the slopes, and one suited to the slopes left the plain almost untouched. Such a survey can be cleaned in two runs. The first run covers the whole survey; besides the slope and centre-beam settings it tightens the blunder test, because the nadir sector flags only shallow soundings by default and the default blunder test rejects only soundings more than 25% of the depth deeper than the predicted surface:

```
mbclean −Idatalist.mb-1 −−cube−spacing=50 −−cube−order=special \
  −−cube−model=depth −−cube−model−a=0.13 −−cube−model−b=0.003 \
  −−cube−sigma−h=1 −−cube−model−c=0 \
  −−cube−prediction=median/4 −−cube−extractor=predicted \
  −−cube−flag=outlier −−cube−flag−threshold=5 \
  −−cube−flag−nadir=10/2 −−cube−blunder=10/0.05/3
```

The second run is limited to the flat area with **−−cube−region**, switches the general outlier test off with a very high threshold, and only removes the crest of shallow soundings at nadir with a low nadir threshold:

```
mbclean −Idatalist.mb-1 −−cube−spacing=50 −−cube−order=special \
  −−cube−model=depth −−cube−model−a=0.13 −−cube−model−b=0.003 \
  −−cube−sigma−h=1 −−cube−model−c=0 \
  −−cube−prediction=median/4 −−cube−extractor=predicted \
  −−cube−region=−11.81/−11.50/36.86/37.14 \
  −−cube−flag=outlier −−cube−flag−threshold=1000 \
  −−cube−flag−nadir=5/0.1
```

The second run applies the edits of the first, so its CUBE surface is built without the soundings already flagged, and its flags are appended to the same edit save files. Run each step once: running the second step again adds more flags. On this survey the two runs flagged 17,953 soundings (0.78%), of which 43% agreed with the manual cleaning, covering 38% of the manual flags; the second run's threshold was chosen by inspecting the result in **mbeditviz**. The region, the sectors and the thresholds belong to this survey. The beam angle is measured from the sonar position of the ping; on this survey the nadir soundings lay several degrees from it along track, so a sector of 5 degrees covered about 3 degrees on either side of nadir. With **−V** the beam angle of each flagged sounding is listed, which shows how the sector falls on other data.

SEE ALSO

mbsystem(1), **mbedit**(1), **mbinfo**(1) **mbprocess**(1), **mbgrid**(1),

BUGS

The algorithms implemented in **mbclean** simply don't detect all bathymetric artifacts that are obvious to the eye on contour charts. Although **mbclean** often does a credible first pass at flagging obvious artifacts, we strongly recommend that any swath bathymetry processing stream include interactive editing of the bathymetry data (e.g. **mbedit**).